

Md. Abdur Rahman

📍 Dhaka, Bangladesh ✉ rahmanmdcs@gmail.com ☎ +8801603482821 in LinkedIn 🐙 Github
 🎓 Google Scholar

Education

Southeast University Oct 2021 – Oct 2025
BSc in Computer Science and Engineering

- **CGPA: 3.87/4.0**
- Relevant Coursework (Grade): Artificial Intelligence (A+), Image Processing (A+), Data Mining (A+), Research Methodology (A+), Algorithms (A+), Advanced Algorithms (A+), Data Structures (A+), Cloud Computing (A+), Probability & Statistics (A+), Calculus (A+), Vector Analysis (A+)

Experience

Research Assistant (Full-time) Gazipur, Bangladesh
Islamic University of Technology (IUT), OIC Nov 2025 – Present

- Architected decentralized Deep Learning and Federated Learning frameworks to address privacy constraints in large-scale data analysis.
- Developed novel algorithms for Causal Inference to improve auditing transparency and predictive accuracy in high-dimensional datasets.
- Authored and submitted three Q1 journal manuscripts proposing new methodologies in decentralized AI and causal inference.

Research Assistant (Part-time) Dhaka, Bangladesh
CAIR, Southeast University Jan 2025 – Oct 2025

- Designed advanced Computer Vision architectures, including Vision Transformers (ViT), Hybrid TransUNet, and Cross-Modal networks, for medical image analysis, integrating XAI methods (SHAP/LIME) and attention mechanisms to provide interpretable clinical decision support.
- Performed high-precision segmentation and classification on complex MRI and X-ray datasets to enhance diagnostic accuracy.
- Achievements: Two papers accepted at IEEE COMPAS 2025; two Q1 journal manuscripts and one additional conference paper currently under review.

Publications

MangoLeafNet-XAI: An Attention-Enhanced Deep Learning Architecture for Accurate and Interpretable Mango Leaf Disease Classification [Paper] [Code]

Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Farzan Majeed Noori, Md Zia Uddin, Abdul Kadar Muhammad Masum
[Frontiers in Plant Science](#)

FedMal-XAI: An Explainable Federated Vision Transformer Leveraging Knowledge Distillation for Privacy-Preserving Malaria Detection [Paper] [Code]

Md. Tofael Ahmed Bhuiyan, *Md. Abdur Rahman*, Md Fokrul Islam Khan, Farzan M. Noori, Abdul Kadar Muhammad Masum
[Frontiers in Public Health](#)

Bridging Data Silos in Corporate Governance: A Hierarchical Stacking Ensemble with Federated Dynamic Aggregation

Abdul Kadar Muhammad Masum, Md. Abul Kalam Azad, Md. Tofael Ahmed Bhuiyan, *Md. Abdur Rahman*
[Applied Computational Intelligence and Soft Computing](#)

AlzFed-XAI: High-Fidelity Interpretable Alzheimer's Diagnosis with Privacy-Preserving Federated Learning [Paper] [Code]

Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Abdul Kadar Muhammad Masum

5th Muslims in ML Workshop co-located with NeurIPS 2025 [🔗](#)

ChakmaBridge: A Five-Way Parallel Corpus for Navigating the Script Divide in an Endangered Language

[Paper] [🔗](#) [Code] [🔗](#)

Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Abdul Kadar Muhammad Masum

Second Bangla Language Processing Workshop co-located with AACL-IJCNLP [🔗](#)

BhasaBodh: Bridging Bangla Dialects and Romanized Forms through Machine Translation

[Paper] [🔗](#) [Code] [🔗](#)

Md. Tofael Ahmed Bhuiyan, *Md. Abdur Rahman*, Abdul Kadar Muhammad Masum

Second Bangla Language Processing Workshop co-located with AACL-IJCNLP [🔗](#)

Detecting Manipulation in Ukrainian Telegram: A Transformer-Based Approach to Technique Classification and Span Identification

[Paper] [🔗](#) [Code] [🔗](#)

Md. Abdur Rahman, Md Ashiqur Rahman

Proceedings of the Fourth Ukrainian Natural Language Processing Workshop (UNLP 2025), Association for Computational Linguistics. [🔗](#)

FAKD-XAI: Feature-Aligned Knowledge Distillation with Explainable AI for Efficient Brain Tumor Classification

[Paper] [🔗](#) [Code] [🔗](#)

Md. Abdur Rahman, Sabik Aftahee, Md Ashiqur Rahman, Lamim Zakir Pronay

International Conference on Data Science, AI and Applications (ICDSAIA) [🔗](#)

MRAViT-XAI: A Novel Multi-Resolution Attention Vision Transformer Framework with Explainable AI for Enhanced Lung and Colon Cancer Classification

[Paper] [🔗](#) [Code] [🔗](#)

Md. Abdur Rahman, Sabik Aftahee, Lamim Zakir Pronay, Md Ashiqur Rahman

IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN) [🔗](#)

PneumoniaXAttnNet: An Attention-Driven Xception Framework for Accurate Pneumonia Classification with XAI

[Paper] [🔗](#)

Tanvir Ahmed, *Md. Abdur Rahman*, Md. Delower Hossain, Yousuf Howlader, Md. Mahi Uddin, Md. Tajuddin, MD Illeas Hossain, Md. Mijanur Rahman

2nd IEEE Conference on Computing Applications and Systems (COMPAS 2025) [🔗](#)

CerviTrans-XAI: An Explainable Vision Transformer Ensemble for Accurate Cervical Cancer Classification

[Paper] [🔗](#)

Md. Delower Hossain, Md. Mijanur Rahman, *Md. Abdur Rahman*, Tanvir Ahmed, Sujay Dhar, Orna Biswas

2nd IEEE Conference on Computing Applications and Systems (COMPAS 2025) [🔗](#)

Advancing Military Cybersecurity: A Scalable Ensemble Network Intrusion Detection System Framework with SHAP Analysis for Military Operations

[Paper] [🔗](#)

Md. Tofael Ahmed Bhuiyan, *Md. Abdur Rahman*, Prof. Dr. Abdul Kadar Muhammad Masum, Arif Hossen, Md Ruhul Rabbi

6th IEEE International Conference on Telecommunications and Photonics (ICTP 2025) [🔗](#)

DeepSea-Net: A YOLO Based Framework for Real-Time Detection and Classification of Underwater Plastic Pollution

[Paper] [🔗](#)

Progga Parmita Roy, Sabbir Sarder Munna, Md. Sazed Ul Karim, *Md. Abdur Rahman*, Md. Tofael Ahmed Bhuiyan, Md. Joseb Siddique Siam

2025 IEEE 7th International Conference on Sustainable Technologies for Industry 5.0 (STI 2025) [🔗](#)

SmolLab_SEU at BEA 2025 Shared Task: A Transformer-Based Framework for Multi-Track Pedagogical Evaluation of AI-Powered Tutors [Paper] [🔗](#) [Code] [🔗](#)
Md. Abdur Rahman, Md. Al Amin, Sabik Aftahee, Muhammad Junayed, Md. Ashiqur Rahman

Proceedings of the 20th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2025), Association for Computational Linguistics. [🔗](#)

SmolLab_SEU at MAHED 2025: Do Arabic-Native Encoders Surpass Multilingual Models in Detecting the Nuances of Hope, Hate, and Emotion? [Paper] [🔗](#) [Code] [🔗](#)

Md. Abdur Rahman, Md Sabbir Dewan, Md. Tofael Ahmed Bhuiyan, Md Ashiqur Rahman

Proceedings of the Third Arabic Natural Language Processing Conference (ArabicNLP 2025), Empirical Methods in Natural Language Processing (EMNLP 2025). [🔗](#)

FedAvg-ResNet50: Privacy-Preserving Explainable Acute Leukemia Classification (Accepted & Presented) [Paper] [🔗](#)

Md. Tofael Ahmed Bhuiyan, *Md. Abdur Rahman*, Zakir Hossain, Md. Fokrul Islam Khan, Abdul Kadar Muhammad Masum

5th International Conference on Applied Intelligence and Informatics (AII 2025, USA) [🔗](#)

X-ArsenicDistill: An Efficient and Interpretable Framework for Arsenicosis Classification via Knowledge Distillation (Accepted & Presented) [Paper] [🔗](#)

Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Mohammad Mahmudur Rahman, Iffat Sania Hossain, Abu Kowshir Bitto, Prof. Dr. Abdul Kadar Muhammad Masum*

28th International Conference on Computer and Information Technology (ICCIT 2025) [🔗](#)

MalariaNet-XAI: An Explainable Deep Ensemble Framework for Accurate Malaria Detection (Accepted & Presented) [Paper] [🔗](#)

Md. Abdur Rahman, Najmus Saadat, Ashfaqu Islam, Khandaker Mohammad Mohi Uddin

28th International Conference on Computer and Information Technology (ICCIT 2025) [🔗](#)

SentimentFusion-Net: A Hybrid ML-DL Framework for Benchmarking ChatGPT User Opinions on Twitter (Accepted & Presented) [Paper] [🔗](#)

Md. Tofael Ahmed Bhuiyan, *Md. Abdur Rahman*, Md. Jobaer Ahmed, Md. Fokrul Islam Khan Khan, Prof. Dr. Abdul Kadar Muhammad Masum

28th International Conference on Computer and Information Technology (ICCIT 2025) [🔗](#)

SmolLab_SEU at CheckThat! 2025: How Well Do Multilingual Transformers Transfer Across News Domains for Cross-lingual Subjectivity Detection? [Paper] [🔗](#) [Code] [🔗](#)

Md. Abdur Rahman, MD AL AMIN, Md Sabbir Dewan, Md Jahid Hasan, Md Ashiqur Rahman

CheckThat! Lab Workshop, 16th Conference and Labs of the Evaluation Forum [🔗](#)

Hinterwelt@LT-EDI 2025: A Transformer-Based Approach for Identifying Racial Hoaxes in Code-Mixed Hindi-English Social Media Narratives [Paper] [🔗](#) [Code] [🔗](#)

Md. Abdur Rahman, MD AL AMIN, Sabik Aftahee, Md Ashiqur Rahman
LT-EDI Workshop, 5th Conference on Language, Data and Knowledge [🔗](#)

Hinterwelt@LT-EDI 2025: A Transformer-Based Detection of Caste and Migration Hate Speech in Tamil Social Media [Paper] [🔗](#) [Code] [🔗](#)

MD AL AMIN*, Sabik Aftahee*, *Md. Abdur Rahman**, Md Sajid Hossain Khan, Md Ashiqur Rahman

LT-EDI Workshop, 5th Conference on Language, Data and Knowledge [🔗](#)

* Authors contributed equally to this work.

Under Review Publications

HA-DynAU-Net: A Hybrid Attention Dynamic Aware U-Net for Precise Skin Lesion Segmentation (Q1 Journal)

1st Author

Submitted to *IEEE Open Journal of the Computer Society*

BEACON: A Novel Cross-Modal Architecture for High-Precision and Explainable Brain Tumor Classification Using Medical-Specific Attention

1st Author

Submitted to *Plos One (Q1 Journal)*

CAMF-Swin: A Hybrid Swin Transformer and Domain-Specific Network with Cross-Modal Attention for Explainable Cervical Cytopathology Classification

1st Author

Submitted to *BMC Women's Health (Q1 Journal)*

HybridTransUNet: Fusing Vision Transformers with Attentive U-Net for Accurate and Interpretable Brain Tumor Segmentation

1st Author

Submitted to *International Journal of Intelligent Systems (Q1 Journal)*

Privacy-Preserving ESG Risk Prediction via Federated Hybrid Neural-Boosting Ensembles

3rd Author

Submitted to *IEEE Access (Q1 Journal)*

Honors & Professional Activities

3rd Position (Poster Presentation), Student Research Day 2025

[Certificate Link](#) 

Champion, Fragments of Feeling: A Contextual Emotion Challenge (2025)

[GitHub Repo](#) 

1st Runners-up, CalorieQuest: Predicting Energy Burn with Data (2025)

[GitHub Repo](#) 

2nd Runners-up, DU AI Challenge (2025)

[GitHub Repo](#) 

Finalist (Top 10), CUET ETE Datathon 2025

[GitHub Repo](#) 

Finalist (Top 7), BUBT Artificial Intelligence Conquest 2025

[GitHub Repo](#) 

Honorable Mention in ICPC Online Preliminary Contest 2024

[Certificate Link](#) 

Reviewer, 20th Workshop on Innovative Use of NLP for Building Educational Applications (BEA), Co-located with 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)

Professional Certification

AWS Certified Cloud Practitioner
Amazon Web Services (AWS)

Dec 2025 - Dec 2028 

GitHub Foundations Certification
GitHub

Nov 2024 - Nov 2027 

Skills

Programming Language: C, C++, Java, Python

Programming Framework: Pandas, NumPy, Matplotlib, Seaborn, PyTorch, HuggingFace, OpenCV, Django, FastAPI, Spring Boot

Technologies: Machine learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), large language model (LLM), Git, GitHub, Microsoft Office, LaTeX